HW#6

Q1. 1D robot — measure squared range + noise

$f_x = 5$ sensor noise $\sim N(0,3)$ odo move noise $\sim N(0,1)$

$t = 0^-$ - turns on around $x_0 \sim N(0,4)$

$t = 0^+$ - observes f_x, measures d^2 or q

$t = 1^-$: moves to x_1, odo measures 1 unit

$t = 2^+$: moves to x_2, odo measures 2 units

$t = 2^+$ from x_2: observe f_x w/ $d^2 = 1$

a) motion model

~~$x_t = x_{t-1} + \varepsilon_t$ $\varepsilon_t \sim N(0,3)$~~ $x_t = x_{t-1} + u_t + \varepsilon_t$ $\varepsilon_t \sim N(0,1)$

b) meas. model

$z_t = (f_x - x_t)^2 + \delta_t$ $\delta_t \sim N(0,3)$

$= (5 - x_t)^2 + \delta_t$

$g(u_t, x_{t-1}) = x_{t-1} + u_t$

$G_t = 1$

$h(x_t) = (5 - x_t)^2$

$H(x_t) = -2(5 - x_t)$

$= 2x_t - 10 \Big|_{x_0 = 0}$

~~$x_t \approx (-10)(x_t) + 10(0) + 25$~~

Prediction:

$\bar{\mu}_0 = 0$ $\bar{\Sigma}_0 = (1)(1)(1) + 4$

$= 5$

correction

$K_0 = (5)(-10)\left((-10)^2(4) + 3\right)^{-1}$

$= 0 - 0.0998$

$\mu_0 = \bar{\mu}_t + K_t\left(z_t - h(\bar{\mu}_t)\right)$

$\mu_0 = 1.588$

$\Sigma_0 = \left(1 + (+0.0998)(-10)\right)(4)$

$\Sigma_0 = 0.0298$

t=1

Prediction

$\bar{\mu}_1 = 2.588$

$\bar{\Sigma}_1 = 0.0298 + 1 = 1.0298$

no corr

correction

$K_t = \frac{(2.0298)(2(4.588)-10)}{(2(4.588)-10)^2(2.0298)+3}$

$= -0.382$

$\mu_2 = \bar{\mu}_2 + K_t(z_t - h(\bar{\mu}_t))$

$= 4.588 + (-0.382)(1-(5-4.588)^2)$

$\mu_2 = 4.271$ ✓

t=2

Prediction

$\bar{\mu}_2 = 4.588$

$\bar{\Sigma}_2 = 2.0298$

$\Sigma_2 = (1-K_2 H_2)\bar{\Sigma}_2 = (1-(-0.382)(2(4.588)-10))(2.0298)$

$\Sigma_2 = 1.391$

$bel(x_2) \sim (4.271, 1.391)$

UKF

$\alpha = 1, \beta = 2$

sig. points

$= \mu, \mu \pm \sqrt{(1+\lambda)\Sigma}$

$= \mu, \mu \pm \sqrt{\Sigma}$

$\lambda = 0$ | ~~$w_m = 0$~~ $w_m^0 = 0$

$w_c^0 = 2$

$w_m^i = w_c^i = \frac{1}{2}$

$\psi^i = g(x^i)$

$\mu' = \sum_{i=0}^{2n} w_m^i \psi^i$

$\Sigma' = \sum_{i=0}^{2n} w_c^i (\psi^i - \mu^i)(\psi^i - \mu')^T$

Pred

$X_0 = \begin{bmatrix} 0 \\ 2 \\ -2 \end{bmatrix}$

~~$\Psi_0 = \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix}$~~

$\bar{\mu}_t = (1)(0) + (3)(\frac{1}{2}) + (-1)(\frac{1}{2})$

$\bar{\mu}_t = (1)$

$\bar{\Sigma}_t = (2)(0)^2 + (\frac{1}{2})(2)^2 + (\frac{1}{2})(2)^2 + 1$

$= 5$

$\bar{\Sigma} = \frac{1}{2}(-2)(20) + \frac{1}{2}(2)(-20) = -40$

$K = \bar{\Sigma} S^{-1}$

$= \frac{-40}{435}$

$= -0.0920$

corr

$x_0 = 0, -2, 2$

$z_t = 25, 49, 9$

$\hat{z}_0 = (0)(25) + \frac{1}{2}(49) + \frac{1}{2}(9) = 29$

$S_0 = 2(4)^2 + (\frac{1}{2})(20)^2 + (\frac{1}{2})(20)^2 + 3 = 435$

$\mu_0 = 0 + (-0.092)(9-29)$

$\mu_0 = 1.839$

$\Sigma_0 = \bar{\Sigma}_0 - K_0 S_0 K_0$

$\approx 440 - (-0.092)^2 (435)$

$\Sigma_0 = 0.322$

$t = 1^-$

$X_{t-1} = (1.839, 2.406, 1.272)$

$\bar{X}_t = (2.839, 3.406, 2.272)$

$\bar{\mu}_t = 2.839$

$\bar{\Sigma}_t = \frac{1}{2}(3.406 - 2.839)^2 \Sigma_0 + R = 0.322 + 1 = 1.322$

$t = 2^-$

$\bar{\mu}_2 = 4.839$

$\bar{\Sigma}_2 = 2.322$

$\bar{Z}_t = \bar{Z}^+$

$\bar{X}_2 = (4.839, 6.363, 3.315)$

$\bar{Z}_2 = (0.026, 1.858, 2.839)$

$\hat{Z}_2 = \frac{1}{2}(1.858) + \frac{1}{2}(2.839) = 2.349$

$\bar{S}_2 = 2(2.349 + 0.026)^2 + \frac{1}{2}\left((1.858 - 2.349)(6.363 - 4.839) + (2.839 - 2.349)(3.315 - 4.839)\right)$

$\bar{S}_2^{12} = -0.748$

$S_2 = 2(0.026 - 2.349)^2 + \frac{1}{2}\left((1.858 - 2.349)^2 + (2.839 - 2.349)^2\right) + 3$

$S_2 = 14.03$

$K_t = \bar{\Sigma}_2 S_2^{-1} = \frac{-0.748}{14.03} = -0.0533$

$\mu_2 = \bar{\mu}_2 + K_2(Z_t - \hat{Z}_t)$

$= 4.839 + (+0.0533)(1 - 2.349)$

$= 4.911$

$\Sigma_2 = 2.322 - (-0.0533)^2(14.03)$

$\Sigma_2 = 2.282$